

Increasing microtubule acetylation rescues axonal transport and locomotor deficits caused by LRRK2 Roc-COR domain mutations

SCIFACT-EVIDENCE-9638032

Evidence abstract

Leucine-rich repeat kinase 2 (LRRK2) mutations are the most common genetic cause of Parkinson's disease.

LRRK2 is a multifunctional protein affecting many cellular processes and has been described to bind microtubules.

Defective microtubule-based axonal transport is hypothesized to contribute to Parkinson's disease, but whether LRRK2 mutations affect this process to mediate pathogenesis is not known.

Here we find that LRRK2 containing pathogenic Roc-COR domain mutations (R1441C, Y1699C) preferentially associates with deacetylated microtubules, and inhibits axonal transport in primary neurons and in *Drosophila*, causing locomotor deficits in vivo.

In vitro, increasing microtubule acetylation using deacetylase inhibitors or the tubulin acetylase TAT1 prevents association of mutant LRRK2 with microtubules, and the deacetylase inhibitor trichostatin A (TSA) restores axonal transport.

In vivo knockdown of the deacetylases HDAC6 and Sirt2, or administration of TSA rescues both axonal transport and locomotor behavior.

Thus, this study reveals a pathogenic mechanism and a potential intervention for Parkinson's disease.

Document metadata

Corpus | SciFact

Document ID | 9638032